Lecture #3: Types of Variables & Conducting a Statistical Study

STAT 200 – Introductory Statistics

Torin Quinlivan

Knox College

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Today's Objective

After today's lecture, you should:

- classify variables as categorical or numeric; as nominal, ordinal, interval, and ratio.
- know the process of conducting a statistical study.
- identify types of observational studies and experimental studies.
- properly use the terminology of studies.
- know the three keys to a successful experiment.
- explain how to protect the experiment and the subject.

Example

Neurologists want to study the effect of Vitamin C on nerve pain. The goal of the study is to see if taking an intravenous dose of Vitamin C will reduce the amount of nerve pain reported by patients.

To achieve this goal, the researchers advertise at their local university for volunteers with nerve pain. Each member of the study is put into one of two groups: control (no Vitamin C) and treatment (5000 mg Vitamin C daily).

At the start of the study, everyone is given a pain-scale analysis (wherein the patient identifies the level of nerve pain from a scale of 0 to 10). At 90 days, everyone repeats the analysis.

Example

Use the previous description and identify the following:

1. the target population.
2. the sampled population.
3. the sample.
4. the variables.

Types of Variables

Variables are at the heart of statistics. They are what we measure on our subjects, and they are what we analyze.
Different types of variables contain different levels of information.

Types of Variables

While there are several ways to categorize variables based on their characteristics, here are three different classification systems to consider:

- qualitative and quantitative
- discrete and continuous
- nominal and ordinal and interval and ratio

All three schemes are useful in helping us better understand what can and cannot be done with certain variables, and what they can describe.

Qualitative vs. Quantitative Data

Qualitative data, also known as **categorical data**, consist of labels or descriptions of traits.
Examples:

- hair color
- phone type
- car make
- government type
- group number

Qualitative vs. Quantitative Data

Quantitative data, also known as **numeric data**, consist of counts or measurements. Examples:

- temperature
- height
- number of pages read in the past week
- number of pizza slices eaten in a month
- grade point average

Qualitative vs. Quantitative Data

Be careful! Just because a variable is a number does ***not*** mean that it is necessarily quantitative.
For instance, your Knox ID is not quantitative data; as an identifier, it is a label for you, thus a qualitative variable.
The way I think about it is if your variable has a unit (inches, pounds, dollars, years, etc.) or if you can take an average of it, it is likely quantitative.
What does it mean to take the average age of everyone in the class? What would it mean to take the average ID number of everyone in the class?

Continuous vs. Discrete Data

Discrete variables are quantitative variables that can take on only particular values and are usually counts.
Examples:

- number of bumps on a log
- number of students in the room
- number of points scored in an NBA game

Continuous vs. Discrete Data

Continuous variables are quantitative variables that can take on any value in a given interval and are usually measurements.
Examples:

- temperature
- height
- weight

The Four Levels of Measurement

The level of measurement of a variable describes the amount of information that variable contains.

- **Nominal** — description
- **Ordinal** — nominal plus ordering
- **Interval** — ordinal plus meaningful (equal) differences between levels
- **Ratio** — interval plus a true zero value

Mnemonic Device: **NOIR** ("black" in French)

The Four Levels of Measurement

The level of measurement of a variable describes the amount of information that variable contains. The four levels are:

- Data at the **nominal** level of measurement are qualitative data consisting of labels or names.
- Data at the **ordinal** level of measurement are qualitative data that can be arranged in a meaningful order, but calculations such as addition or division do not make sense.
- Data at the **interval** level of measurement are quantitative data that can be arranged in a meaningful order, and differences between data entries are meaningful.
- Data at the **ratio** level of measurement are quantitative data that can be ordered, differences between data entries are meaningful, and the zero point indicates the absence of something.

The Four Levels of Measurement

Nominal	**Ordinal**	**Interval**	**Ratio**
Qualitative	Qualitative	Quantitative	Quantitative
	Order is meaningful	Order is meaningful	Order is meaningful
		Differences between values are meaningful	Differences between values are meaningful
			Zero indicates absence

Examples

Identify each of the following as **quantitative** or **qualitative** (numeric or categorical):

- the temperature outside on a Saturday
- whether the student wears a mask or not
- the amount of money carried by the person
- the number of credit cards carried by a person
- your favorite grocery store in Galesburg
- the number of allergies in a person

Examples

Identify each of the following as **continuous** or **discrete**:

- the temperature outside on a Saturday
- whether the student wears a mask or not
- the amount of money carried by the person
- the number of credit cards carried by a person
- your favorite grocery store in Galesburg
- the number of allergies in a person

Examples

Identify the level of each of the following variables (**nominal**, **ordinal**, **interval**, or **ratio**):

- the temperature outside on a Saturday
- whether the student wears a mask or not
- the amount of money carried by the person
- the number of credit cards carried by a person
- your favorite grocery store in Galesburg
- the number of allergies in a person

Conducting a Statistical Study

These are the general steps to conduct a **legitimate** scientific study:

1. Determine the design of the study.
   a. State the question to be studied.
   b. Determine the population and variables.
   c. Determine the sampling method.
2. Collect the data.
3. Organize the data.
4. Analyze the data to answer the question.

Conducting a Statistical Study

There are two basic types of scientific studies. They are:

- An **observational study** observes data that already exist.
- An **experiment** generates data to help identify cause-and-effect relationships.

An observational study takes the subjects as they are. In an experiment, generally, the researchers assign the subjects to groups randomly.

The Goal of the Sample

There is only one purpose/goal of the sample: It must be **representative** of the population.
A **representative sample** has the same relevant characteristics as the population and does not favor one group from the population over another.
Note: A sample can be representative for one characteristic of the population (parameter) but not for another. In fact, it can be representative in every single parameter, but not for the relationships between them.
How does one know if a sample is representative of the population?

Types of Observational Studies

Broadly speaking, a statistical study is either experimental or observational. There are a few main types of observational studies:
In a **cross-sectional study**, data are collected at a single point in time.
In a **time-series study**, data are gathered by following a single person over a period of time.
In a **longitudinal study**, data are gathered by following a particular group over a period of time.
In a **cross-sectional-time-series study**, data are gathered by following a larger group over a period of time.

Terminology of Experimental Studies

Here is some terminology related to scientific experiments. Some of these we have already used, others not.
The **subjects** (**participants**) are people or things being studied in an experiment.
A **treatment** is some condition that is applied to a group of subjects in an experiment.

Terminology of Experimental Studies

Here is some terminology related to scientific experiments. Some of these we have already used, others not.

The **response variable** is the variable in an experiment that responds to the treatment. This type of variable is also termed the **dependent** variable.

The **explanatory variable(s)** is the variable (or variables) in an experiment that is thought to cause the change in the response variable. This type of variable is also termed the **independent** variable.

Terminology of Experimental Studies

Here is some terminology related to scientific experiments. Some of these we have already used, others not.

A **control group** is a group of subjects to which no treatment is applied in an experiment.

- A **placebo** is a substance that appears identical to the actual treatment but contains no intrinsic beneficial elements.
- The **placebo effect** is a response to the power of suggestion, rather than the treatment itself, by participants of an experiment.

Terminology of Experimental Studies

Here is some terminology related to scientific experiments. Some of these we have already used, others not.

A **treatment group** is a group of subjects to which researchers apply a treatment in an experiment.

Confounding variables are factors other than the treatment that are related to both the explanatory and the response variable, so that their effects cannot be separated from the effect of the treatment.

The Fundamental Principle of Experiments

To properly perform a scientific experiment, one should:

1. completely randomize the control and treatment groups.
   - or, for reasons that improve the statistics (a matched-pairs design), explicitly tie each member of the control group to a member of the treatment group
2. control for outside effects on the response variable.
3. analyze a sufficiently large sample.
4. replicate the experiment a significant number of times to see meaningful patterns.

For more information on matching (in case-control studies): Please read the two-page article "Epidemiology in Practice: Case-Control Studies" (Lewallen and Courtright, 1998) located at `https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1706071/`.

The Three Keys for a Successful Experiment

There are reasons behind these fundamental principles. They delve into the goal of ensuring the sample is representative of the population.

- The purpose of the above principles is to ensure that any difference in the response variable is due *only* to the treatment (explanatory variable) and *not* to differences in the groups.
- One needs to have a sufficiently large sample size to detect the "signal" and not be confused by the "noise".
- Repeat, using proper sampling methods, to help ensure the average of the results is close to the parameter of interest.

Protecting the Experiment

How can we ensure experiments are properly performed?
In a **single-blind experiment**, subjects do not know if they are in the control group or the treatment group, but the people interacting with the subjects in the experiment know in which group each subject has been placed.
In a **double-blind experiment**, neither the subjects nor the people interacting with the subjects know to which group each subject belongs.

Protecting the Subject

An **Institutional Review Board** is a group of people who review the design of a study to make sure that it is appropriate and that no unnecessary harm will come to the subjects.

The Institutional Review Board (IRB) at Knox College is responsible for overseeing all research at Knox College that involves the use of human subjects. The ultimate goal of the IRB is to safeguard the well-being of individuals who participate in research conducted by Knox College and/or its faculty, staff, and students and to ensure that research occurring at Knox College meets accepted ethical standards. The Knox IRB also reviews projects conducted by outside researchers who wish to collect data from members of the Knox College community.

This information — and more — about the Knox College IRB can be found at `https://www.knox.edu/offices/academic-affairs/institutional-review-board`

Protecting the Subject

Informed consent involves completely disclosing to participants the goals and procedures involved in a study and obtaining their agreement to participate.